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An Introduction to Big Data



An Introduction to Big Data Executive Summary

Theme

Objective

Outcome

1

The four Vs

Describe the characteristics of big data.

- Know the four Vs.

2

Data sources and types

List different types of data sources, and describe the types of data.

- Recognize different data sources (internal/external).
- Know the difference between structured/unstructured/semi-structured data.

3

Big data integration

List the challenges faced with big data engineering and storage.

- Know the considerations that have to be made when architecting a big data solution.

4

Value of big data

Be able to discuss the benefits of big data for business and industry.

- Be able to articulate a big data solution in terms of business value delivered.

What is big data?

Datasets become so large and complex that it is near impossible to process them with traditional techniques.

Data is **constantly growing** as a result of the increase in data-producing sources, including **social media platforms, Internet of Things devices, smartphones, and wearable electronics**. The increase in the size and types of data available creates a demand for new ways to manage, centralize, organize, and utilize data.

Big data is the field that deals with these data-related challenges. This includes ways to **clean, process, analyze,** and **consume** datasets that are **too large for traditional data-processing methods** to deal with.

Challenges most often faced when dealing with big data include:

- Capture
- Search
- Analysis
- Curation
- Sharing
- Visualization
- Storage
- Transfer



Characteristics of big data: Volume

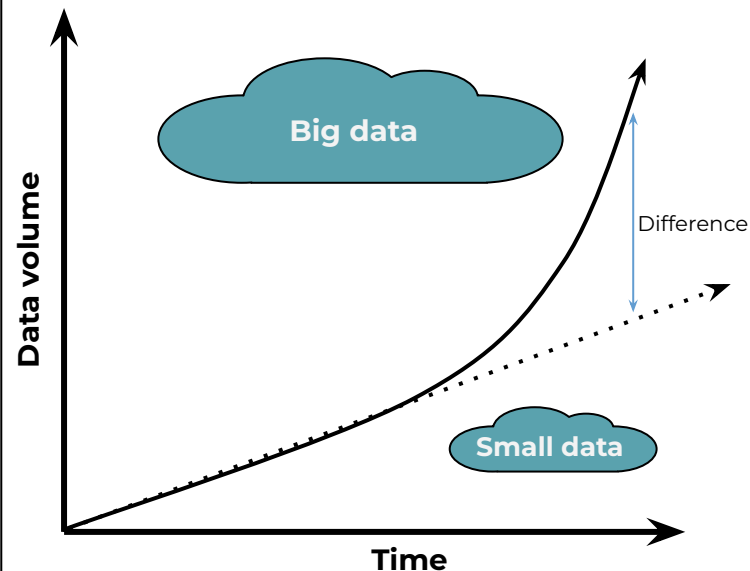
Data volume concerns the sheer amount or size of data. Here we refer to a data collection's actual size, i.e. gigabytes, terabytes, petabytes, etc.

Data volume refers to the **effective size of a collection of files**, typically arranged into a dataset that **needs to be analyzed**. Nowadays, these **collections can be in the order of petabytes or greater**.

We often consider such a collection as **'big data'** when its **volume is too large for traditional data-processing methods to feasibly cope**.

When this is the case, practitioners **resort to alternative methods of data processing**, such as distributed computing, cloud computing, and data partitioning.

Examples of high-volume datasets include all the geospatial, medical, and personal information of the daily COVID-19 cases worldwide.



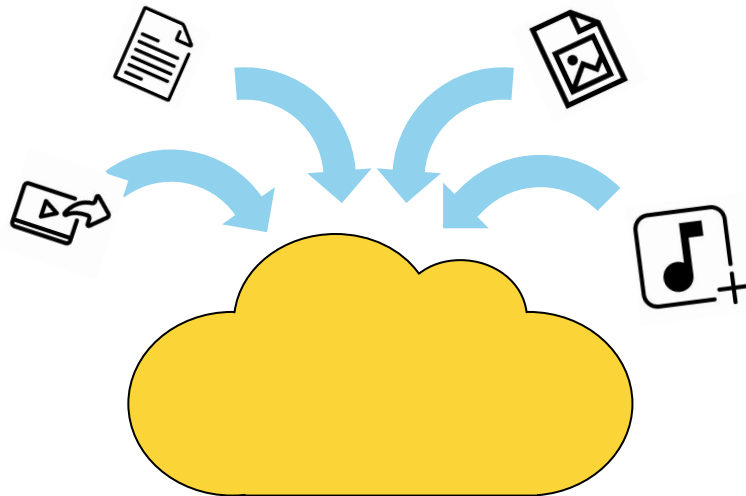
Characteristics of big data: Velocity

Data **velocity** refers to the speed with which data are generated or need to be processed.

Velocity is the frequency of incoming data. In simpler terms, data velocity is the **rate** at which **data** are being **generated** and sourced from applications, platforms, websites, etc.

An example of data that are generated with high velocity would be WhatsApp messages or Instagram posts or financial data from a stock exchange.

A streaming application like Amazon Web Services Kinesis is an example of an application that handles the velocity of data.



Characteristics of big data: Variety

Data **variety** describes the wide range of structures and file types that may need to be processed by our applications.

A good way to visualize the variety of big data is to think of social media sites. They consist of text, audio, and video data streams that need to be processed.

When we think about varieties of data, it is helpful to also think of it as **either structured or unstructured**.

Structured data is the kind of data we find in a fixed field in a record, file, or spreadsheet.

Unstructured data refers to data that doesn't follow a specific format. Think satellite images, photographs and video, social media data, and medical records.

Streaming data adds another layer of complexity, as you can only scan the data once before more up-to-date data flow in.



Characteristics of big data: Veracity

Data veracity refers to the trustworthiness of data.

Veracity refers to the **quality of the data that are being analysed**. When speaking of data veracity, we specifically refer to factors that influence the **'truthfulness' of a given dataset**.

A high-veracity dataset can be regarded as having reputable data and its analysis produces reliable insights and observations.

Datasets with low veracity contain multiple entries that are affected by multiple sources of noise. As such, the results of analyses and insights gained from low-veracity datasets garner little trust.

Things to consider when determining the veracity of data include its **origin, authenticity, trustworthiness, completeness, and integrity**.

An example of a high-veracity dataset would be data from a medical experiment or trial.

Sources of Data Veracity



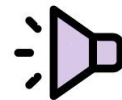
Statistical biases



Lack of data lineage



Software bugs



Noise



Abnormalities



Information Security



Untrustworthy data sources



Falsification



Uncertainty and ambiguity of data



Duplication of data



Out of date and obsolete data



Human error

Where is all of this data coming from?

Short answer: **YOU**



In the past, content was generated by a few companies and consumed by everyone.



Now, everyone generates content using services owned by a few companies, which is then consumed by everyone.

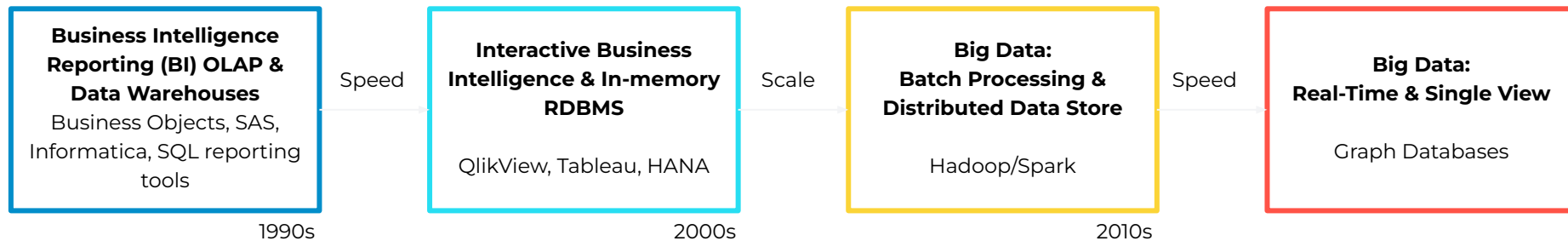


What is driving big data?



Everyone needs an edge, be it for commercial, governmental, military, or social reasons.

The evolution of data flows within business environments



In the past, business intelligence revolved around ad-hoc querying and reporting procedures, manual data-mining techniques, small to medium-sized datasets, and structured data.

Modern businesses have improved their approach to data by optimizing their data workflows, allowing them to process massive amounts of both structured and unstructured data, sometimes even in real-time. This additional processing power also allows them to apply data science techniques to perform predictive analytics.

Big data engineer vs. data scientist



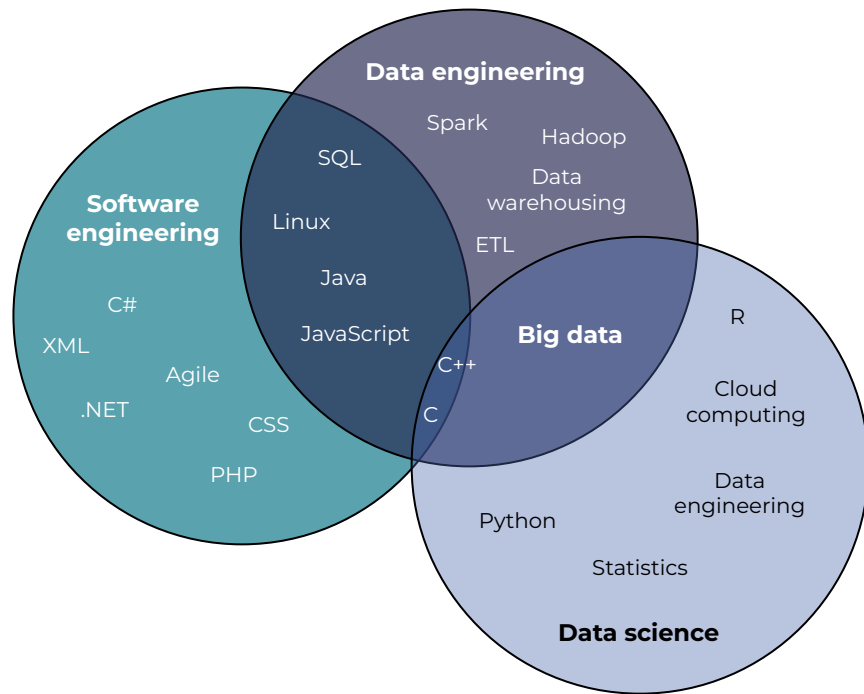
Data engineers are the plumbers building a data pipeline, while data scientists are the painters or storytellers giving meaning to an otherwise static entity.

At a high level, data engineers can be seen as the developers and maintainers of data pipelines and systems used by data scientists. These pipelines support activities such as exploratory data analysis, building and testing models, and using data to test or confirm hypotheses.

More formally defined:

Data engineers are concerned with scoping and developing data pipelines, setting up data models, and developing the end-to-end extract, transform, and load procedures.

Data scientists are concerned with utilizing structured and unstructured data to derive insights and build, deploy, and maintain predictive and prescriptive models to unlock business value.



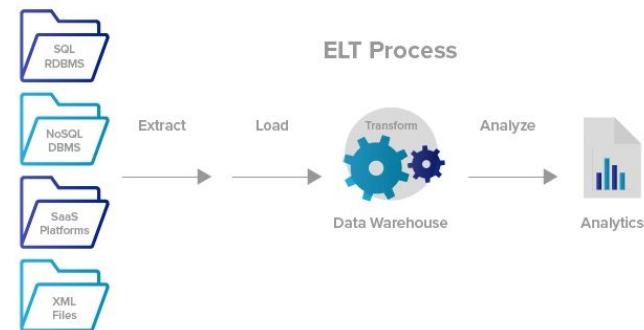
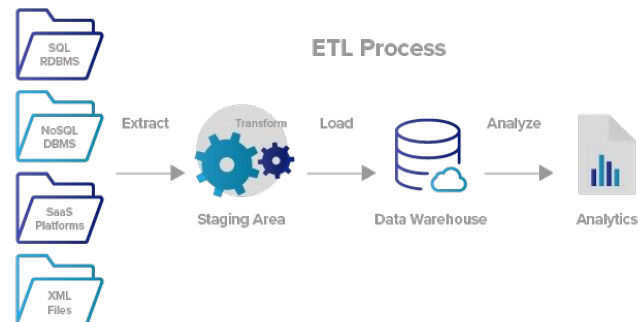
Mapping the flow of big data

How data gets from A to B.

Data that are sourced from websites, applications, IoT devices, etc. are rarely in the desired format. There are usually intermediate steps required to clean, centralize, enrich, and transform these raw data sources. It is here where data engineers make use of an ETL (extract, transform, load) or ELT (extract, load, transform) pipeline.

At a high level, ETL or ELT involves the following steps:

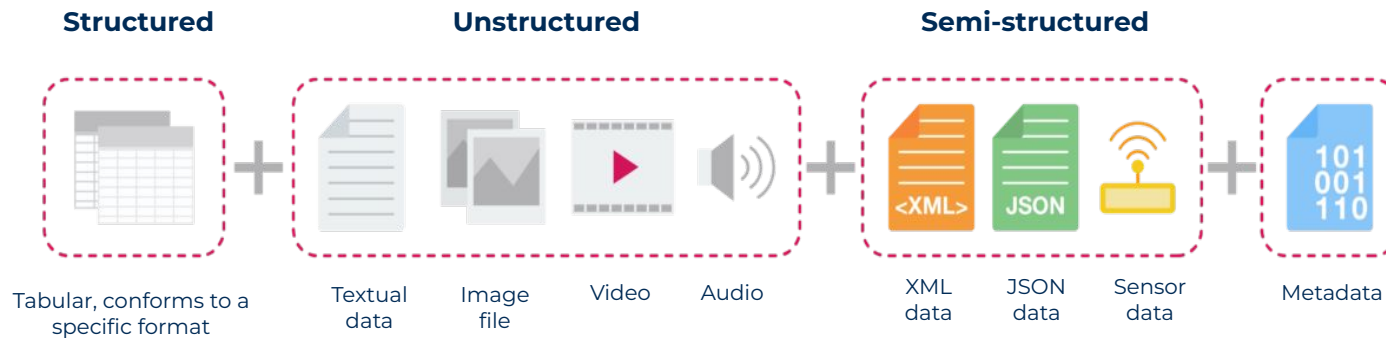
1. **Extract:** With ETL, extract is when we pull the data from the source and dump it into a staging environment. With ELT, the data is moved from the source directly into data storage.
2. **Transform:** Here, transformations are applied to the data so that the data format matches that of the target system.
3. **Load:** With load, the extracted and transformed data are uploaded to the intended data storage system.



Types of big data

Structured, unstructured, and semi-structured.

A data structure is a standard format used to represent, organize, and store data. This data structure will typically include the data values and the relationship between them. A data structure simplifies the access and management of structured and semi-structured data. For the analysis of big data, it is important to understand that there are three common types of data structures:



Note: Metadata is another data category often dealt with in big data engineering. Metadata are data about data, often in a structured format. It provides additional information about a specific set of data. Metadata are frequently used by big data solutions for initial analysis.

Types of big data

Structured, unstructured, and semi-structured.



Structured data: Structured data are organized and consistent for all records within a dataset. These datasets conform to a pre-defined structure and communicate information in an organized manner. An example of a structured dataset includes a SQL database or Excel spreadsheet.



Semi-structured data: Semi-structured data are data that are loosely organized across records in the dataset. This form of data will have (randomly ordered) categories or tags for data to be stored in, but the data within these tags will have no inherent structure. Examples include XML and JSON files.



Unstructured data: Unstructured data are unorganized and inconsistent across the records within a dataset. Examples include text extracted from PDF documents, images, and videos. Unstructured data are by far the most common type of big data processed today (using machine learning) and are more difficult to analyze using traditional software than structured data. Unstructured data can often be stored in a NoSQL database.

Big data storage is challenging



These are the problems to consider and overcome when storing big data.



Infrastructure: If we plan on storing vast amounts of data, we'll need the necessary infrastructure to store it. Solutions include on-prem servers or cloud storage solutions (AWS, Azure, IBM, Google Cloud Platform).



Cost: Depending on what type of data we're storing and how much of it we're storing, we may have to pay for physical hardware and maintenance (on-prem, capital expenditure) or monthly fees for our cloud hosting and storage (off-prem, operating expenses).



UI and accessibility: Data are no good if no one can get to it. We need to ensure that all necessary parts of the business can easily interact and share data (that they are allowed to share).



Security: We should always consider data encryption and limit our reliance on third-party providers.

Big data storage is challenging (continued)



These are the problems to consider and overcome when storing big data.



Corruption: Physical hard drives fail (more often than you think). Therefore, we always need multiple backups (redundancy) for our data.



Scale: We need to ensure that our infrastructure and storage solution (on-prem or off-prem) are capable of easily scaling in the future.



Compatibility: There are many sources of data and data types. Our storage solution must be able to ingest and store any of our business data that may be needed now or in the future.



Talent: There is a scarcity of data professionals worldwide, and the demand for talent is continuously increasing as more companies are trying to gain a competitive edge by leveraging their data.

Data storage technologies: Overview



To overcome the challenges faced when trying to store big data, specialized data-storage technologies are built to deal with the data explosion.

These technologies include big data applications that have moved away from traditional relational database models to processes capable of handling very large amounts of data. These technologies include **distributed file systems, NoSQL databases, SQL databases, and cloud storage solutions.**

In big data processing, we often use distributed computing (more than one computer working together to solve the problem). **Distributed file systems (DFS)** are the storage systems for these distributed computers, which allow these computers (also known as nodes within a cluster) to access shared storage simultaneously. This allows them to each process a small chunk of a large dataset at the same time. A DFS will also duplicate the chunks of data across multiple nodes to achieve both fault tolerance and a higher data throughput. Another word for “data throughput” is **concurrency**, and it refers to the ability of different nodes to process the chunks of the dataset out of order or in partial order, without affecting the final outcome. One of the most often used DFS is the Hadoop Distributed File System (HDFS) by the Apache Software Foundation.



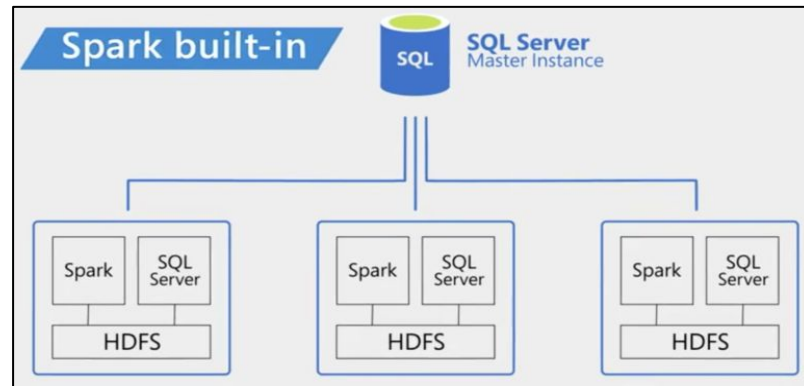
Data storage technologies: SQL



SQL, or Structured Query Language, is one of the most versatile tools that a data professional can use when working with relational databases. Relational data are defined at the basic level by a series of table entities containing columns and rows and linked to other table entities by shared attributes.

The basic unit of the SQL operation is the SQL query. These are question-answer pairs from the designated structured database. A typical SQL query in plain English might be: What is the total number of potato sales in South Africa, in the year 2020 during the month of July? For more on the basics of SQL, read [this](#) blog post.

SQL is a universally accepted and standard language that has been used for decades. Modern architectures still use this language every day to query big datasets. It is now possible to query data stored on multiple servers in record time using distributed storage systems such as Spark.



NoSQL databases are purpose-built for specific data models and have flexible schemas for building modern applications.

NoSQL databases are databases that do not store data in a relation format. Rather, these databases provide some flexibility in the way that a data model can be represented. The most common NoSQL data models include document, key-value pair, column, and graph types. NoSQL databases are preferred to relational databases when working with high-throughput data of large scale with the need for efficient data schema adjustments as the data evolves.

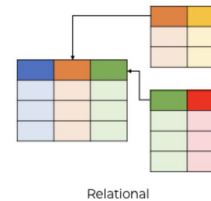
Benefits of NoSQL databases are:

- Flexibility
- Scalability
- High performance
- Highly functional

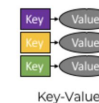
Types of NoSQL databases include:

- Key-value
- Document
- Graph
- In-memory
- Search

SQL DATABASES



NoSQL DATABASES



Data storage technologies: Cloud storage solutions



Modern cloud computing platforms have developed their own solutions to storing, loading, and transforming big data. For enterprises, cloud storage provides flexible access from multiple locations and quick and easy scale capacity (elasticity). Solutions on all platforms allow for managing big data storage and analytics.

Google Cloud



Azure



AWS



Before we get caught up in the technologies involved in big data, we need to take a step back and determine the value our solutions will provide. Big data are meaningless if it does not provide value (considered by many to be the fifth V of big data). Examples of sectors where big data provides value include:

- Healthcare: Patient predictions for improved staffing, electronic health records, real-time alerting, telemedicine, improved supply chain management.
- Public sector: Open government and data sharing, citizen sentiment analysis, economic analysis, smart cities.
- Finance and insurance sectors: Enhanced fraud detection and prevention capabilities, enhanced levels of customer insight, engagement and experience, and enhanced security.
- Energy and transport sectors: Decision support and automation, scalable advanced analytics, increased operational efficiency, data access, exchange, and sharing.
- Media and entertainment sectors: Predict audience interests, provide insights into customer churn, improve customer experience.

Overview of the introduction to big data train:

